Initial Exploratory Analysis English and Urdu

1. Critical drop-offs (leaving questions unanswered or wrong) occur early in English (39% quit before Q7), disproportionately affecting rural students and early grades. https://docs.google.com/document/d/1GtPCdk2CVDD7OeGG3IU3XMDdcQrzBTKwjlpeqXY3Zko/edit?tab=t.9xcldtsnjmn6#bookmark=id.fku7t8w64ixm
2. KMeans clustering revealed three distinct student groups: "Strugglers" (Class 1-2, rural; needing foundational support), "Middle Band" (Class 3-4; poised for growth), and "High Achievers" (Class 5 urban; ready for enrichment).The intervention successfully narrowed achievement gaps—lowest performers (Q1) gained most (+4.2 English/+5.7 Urdu)—but high performers (Q4) declined, indicating insufficient challenge (ceiling effect)(However, this conclusion is less certain for Classes 1 and 2 due to discrepancies between their baseline and endline scores, although quartile-based comparisons remain valid since they are calculated within each class.Link). https://docs.google.com/document/d/1GtPCdk2CVDD7OeGG3IU3XMDdcQrzBTKwjlpeqXY3Zko/edit?tab=t.9xcldtsnjmn6#bookmark=id.9awr5ccjoelb

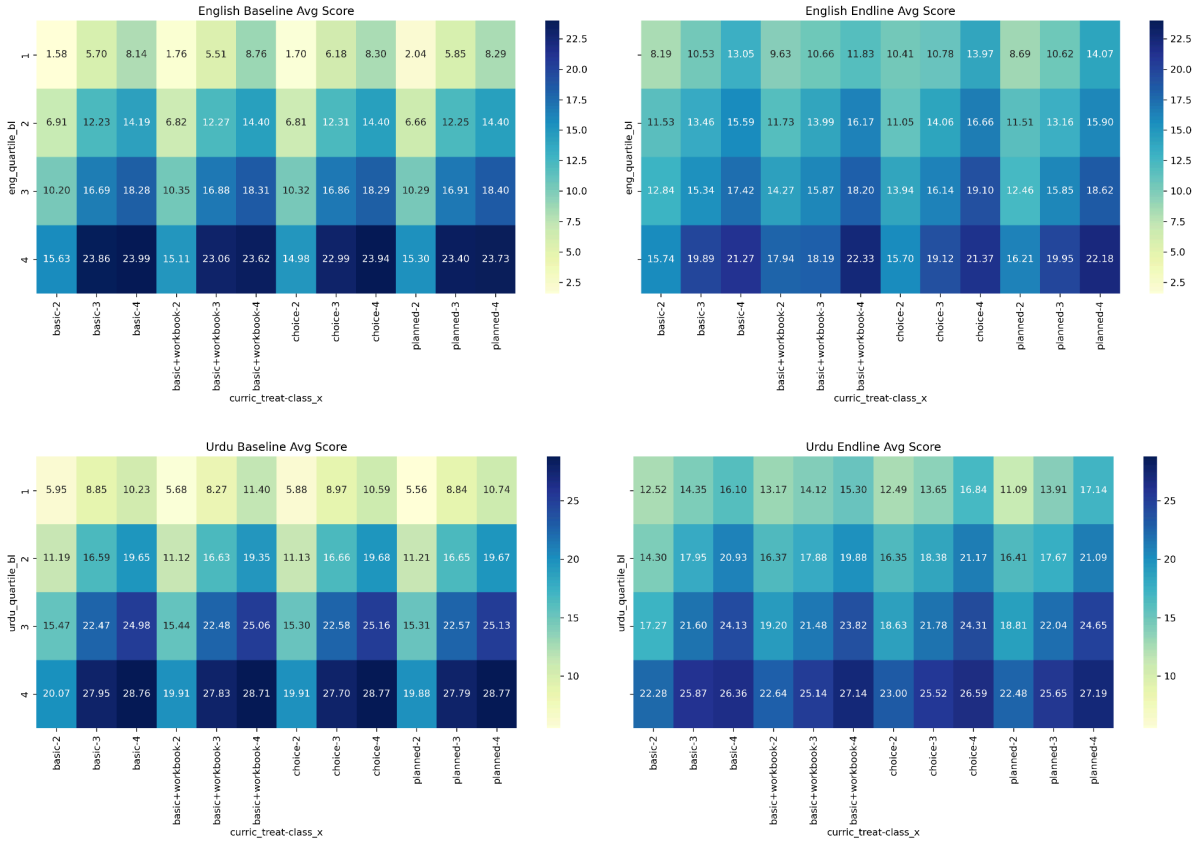
English and Urdu Merging

The process involved cleaning and standardizing tip_ids (removing invalid values, stripping whitespace), dropping duplicates, and performing an inner join between English and Urdu baselines to get 36,763 matches, then recovering 36,769 more via a composite match_key (entry_start_time, username, EMISID, child_id, class_id, section, SchoolName), resulting in 68,048 rows with 89 columns, no duplicate IDs, and an improved merge rate from ~47.6% to 88.2%. A new tip_id column was built from non-missing IDs across baselines, and a corrected class_id resolved 64 mismatches using maximum score rules for Class 1 and 2. Merging the treatment teacher roster with the endline dataset (one-to-many) produced 70,189 rows, reduced to 48,110 valid observations after dropping invalid treatments, with scores standardized and treatments categorized as basic or advanced. For baseline–endline merging, an initial roster-based match reduced observations, so the final approach matched directly on tip_id, yielding 31,947 matches—up from 17,768 possible with only English baseline tip_ids—by first merging English and Urdu baselines via match_key to complete the English tip_id column, likely resolving an issue noted in the Data Handover document. Link.

EDA and Visualization of English and Urdu Dataset

The Choice Treatment is most effective overall, delivering the strongest English gains (+2.26), benefiting both low (Q1) and high performers (Q4), and promoting equity and excellence. Planned Treatment shows balanced retention across middle quartiles (Q2–Q3) with fewer equity drops, while Basic and Basic+Workbook excel for Q1 learners but cause stagnation or negative gains for Q4, especially in Classes 3–5, creating the largest equity gaps. Grade-specific treatments work best when matched to their target class
Link to Summary (However, this conclusion is less certain for Classes 1 and 2 due to discrepancies between their baseline and endline scores, although quartile-based comparisons remain valid since they are calculated within each class. Link.)

Equity Analysis of English Urdu



I found these graphs particularly interesting. The color intensity in the heatmap represents the scores. They display quartiles with average baseline and endline scores

for both English and Urdu side by side. What I observe is that the contrast between the lower and upper quartiles has softened — the colors are blending more, suggesting that the treatments are reducing disparities between quartiles. It could be because of the following reasons:

Spillover effects (cross-subject influence)

- Students might apply problem-solving skills or improved study habits learned during math sessions to other subjects, even if no direct instruction happened in those subjects.

- Teachers involved in the math intervention could unconsciously change their teaching approach for other subjects as well.

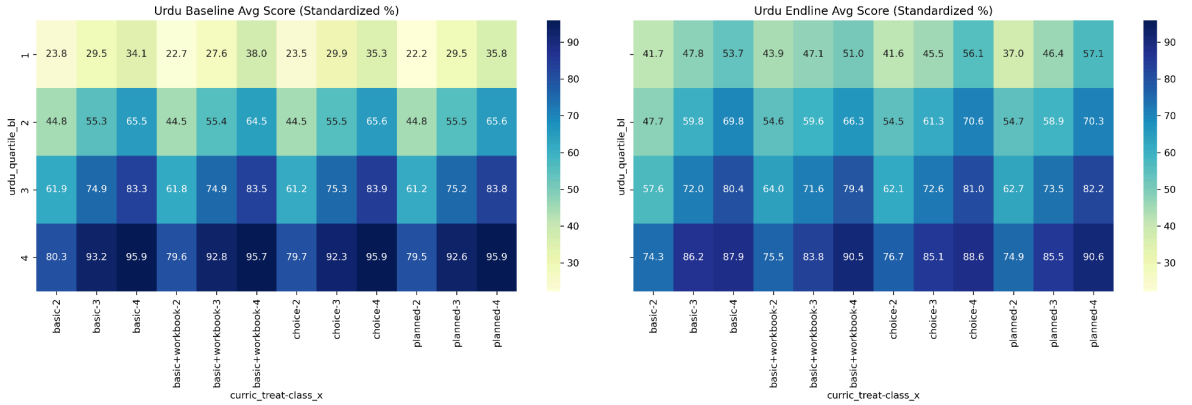
2. Teacher-level treatment diffusion

- If the same teacher teaches both math and other subjects, their improved methods or motivation may carry over to non-math lessons.

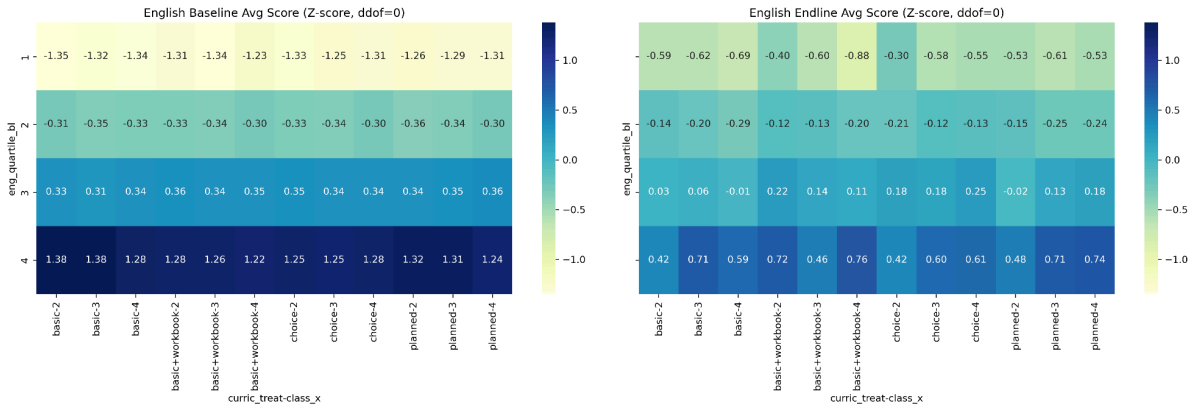
However, there appears to be a ceiling effect, where students in the highest quartile show lower scores in the endline exam. While lower-quartile students benefit, upper-quartile students experience a drop in performance.

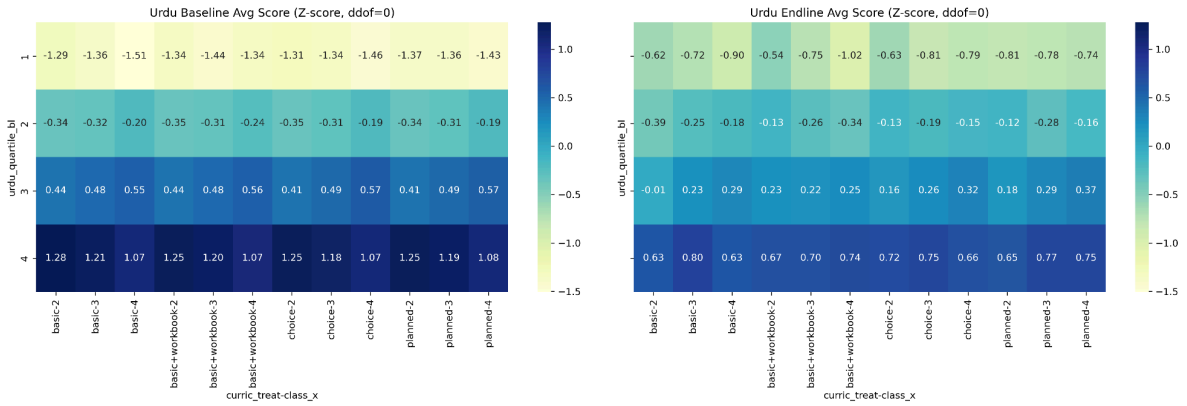
Because the differences in the endline and baseline scores were discovered towards the end , the initial EDA and visualizations were done with raw score so any interpretation and conclusion made on raw endline and baseline score or gain should be taken with caution. I however regenerated the heatmaps using the following code. One with maximum score standardization(Link):

Producing these heatmaps :

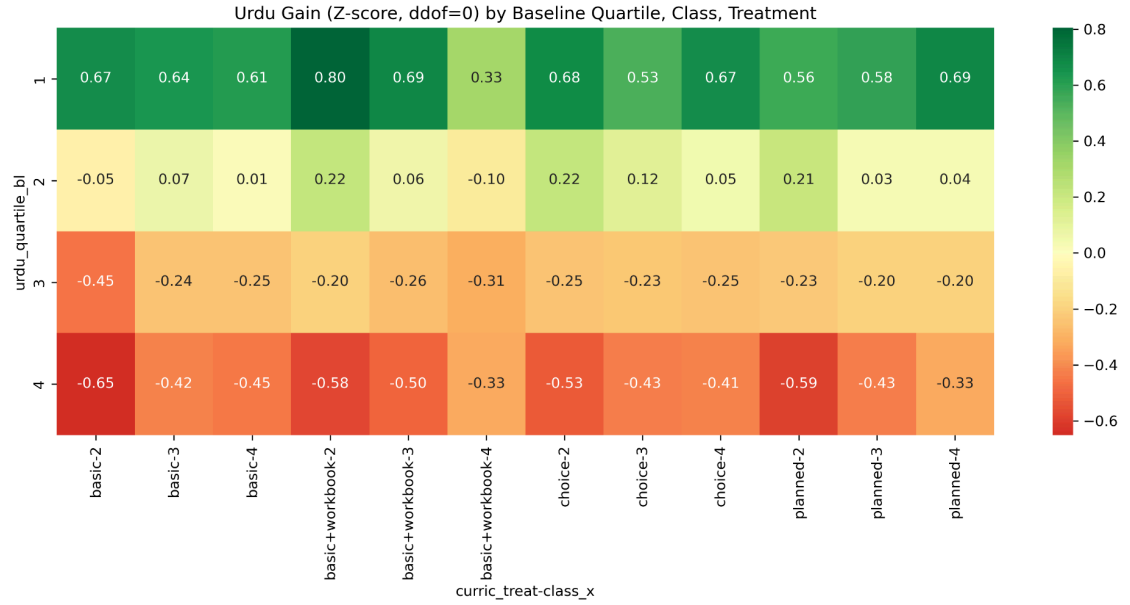


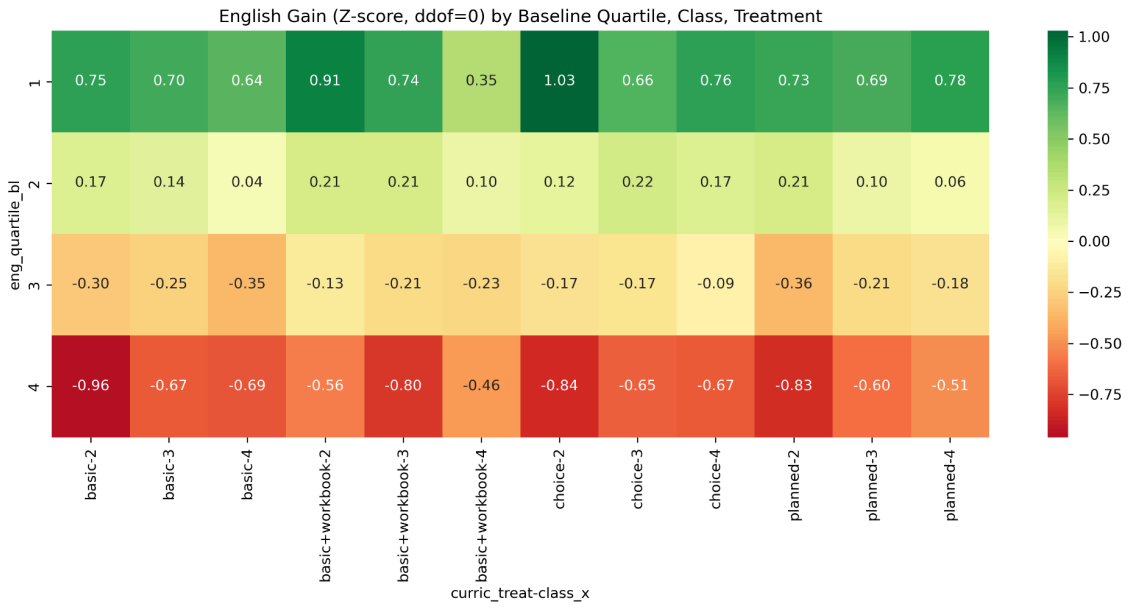
They show a similar color distribution as the raw score. Then I used z-score standardization, It generated the following graphs :





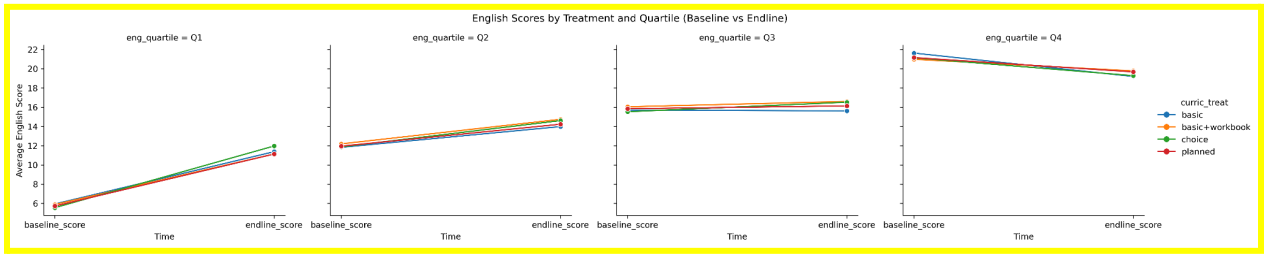
This shows very clear divisions between quartiles . Quartiles 1 ans 2 benefit but quartiles 3 and 4 don't benefit . The within class difference is less profound.
I also generated heatmaps with gains using the z-score standardization:

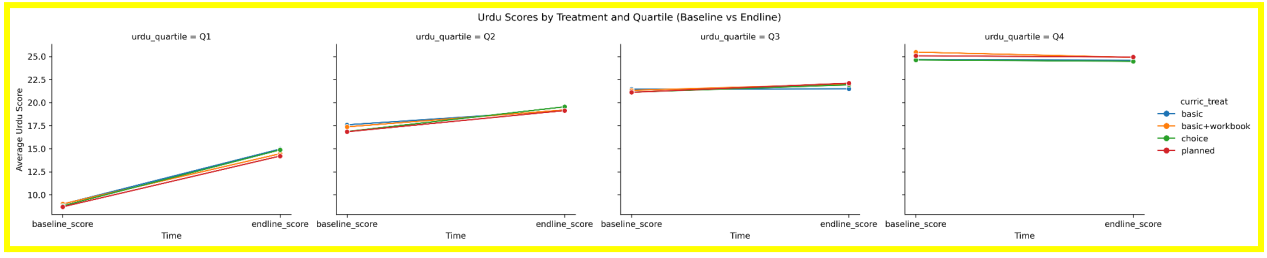


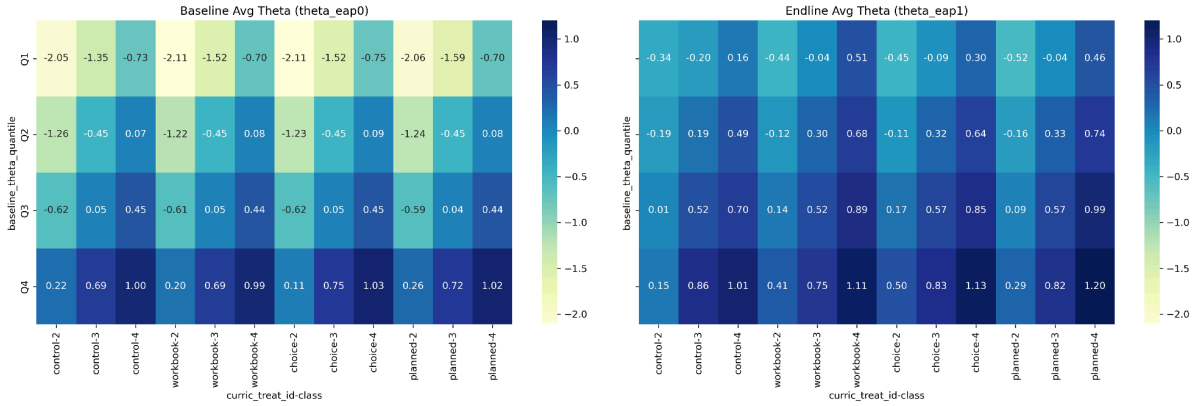


What I observe is the disparity between the quartiles . The disparity is most pronounced in class 2 as seen by the sharp contrast in shades.

I also showed this ceiling effect with line plots (not satndardized)

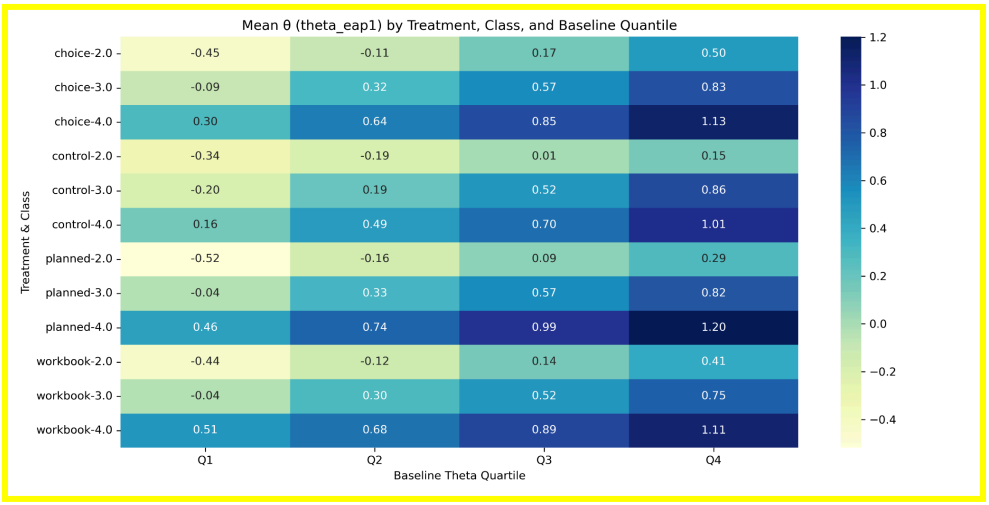


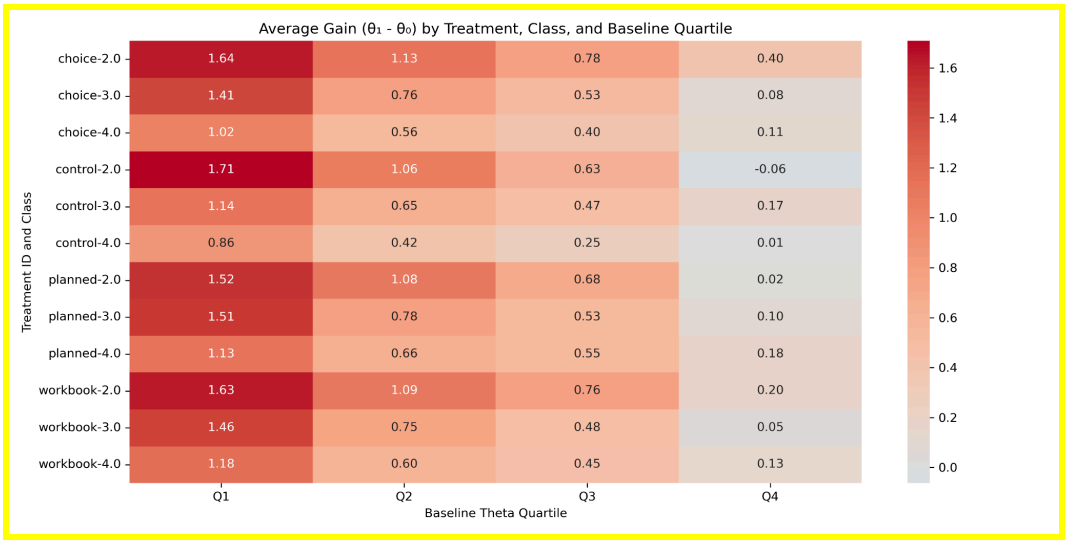




I observe a similar pattern in the math dataset; however, unlike English and Urdu, Quartile 4 shows a positive gain—albeit smaller than Quartile 1 due to a ceiling effect—indicating improved performance. One possible explanation is that the advanced math treatment program benefited even the upper quartiles, but may have diverted their attention from other subjects, resulting in negative gains there. However, since the same downward trend in English and Urdu is also visible in the control group, it is unlikely that the math-specific treatment alone caused this. Other possible explanations include the fact that achieving higher scores in English and Urdu is inherently more difficult than in math, making incremental improvements in math easier. Additionally, if the control group was in the same school, spillover effects from the treatment could also have occurred Link.

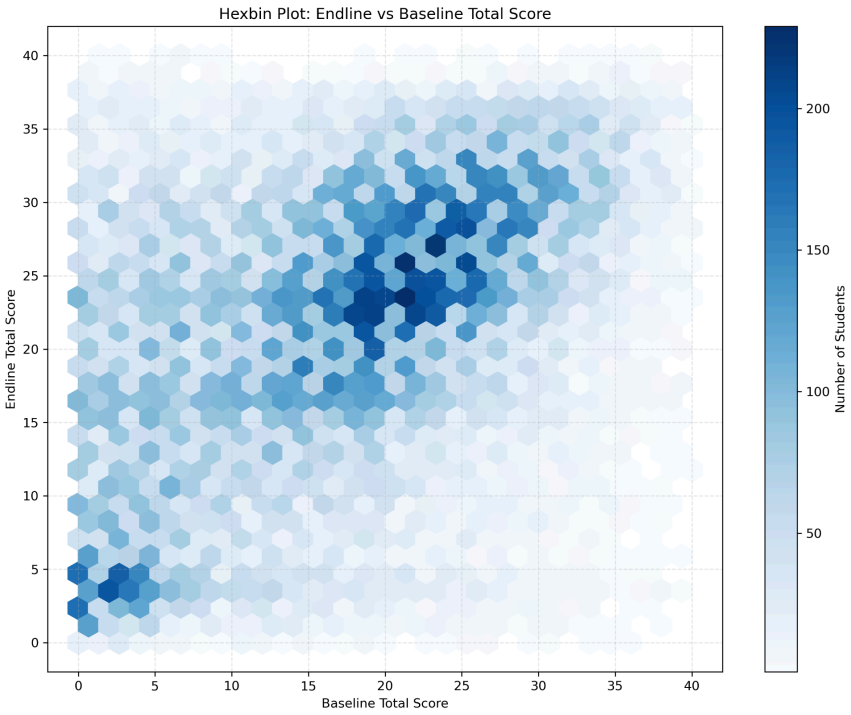
Math Exploration (Equity Analysis)





Looking at the Math Equity Analysis, I see the same trends observed in the English and Urdu Dataset. Although the endline scores for upper quartiles and classes is higher, the gains are lower or even negative when compared to lower quartiles.

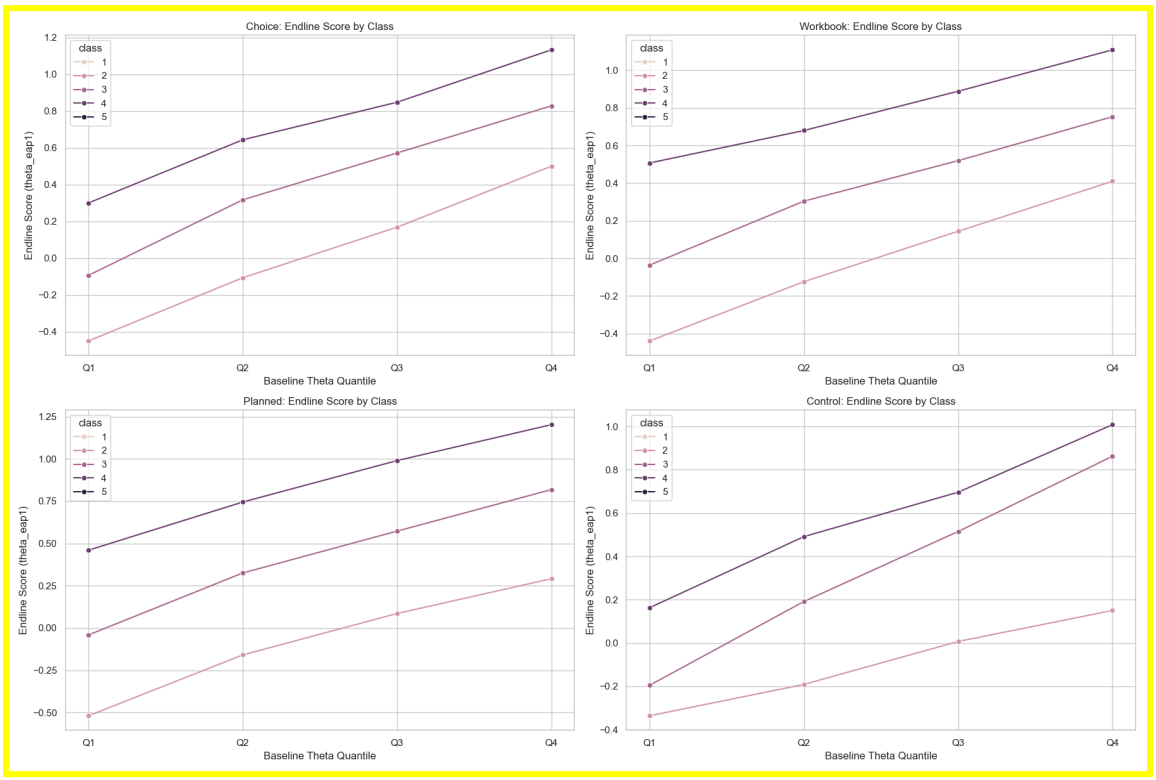
Scatter Plot Math Dataset Baseline Vs Endline(Link to more Scatterplots)



The scatter plots show a linear relation between the Endline and Baseline Scores (Raw not theta).

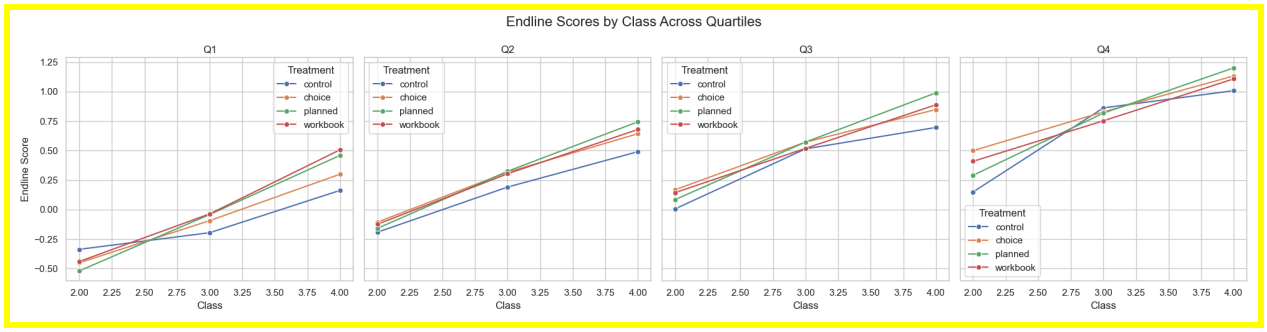
Math Line Plots(Using theta Data Link)

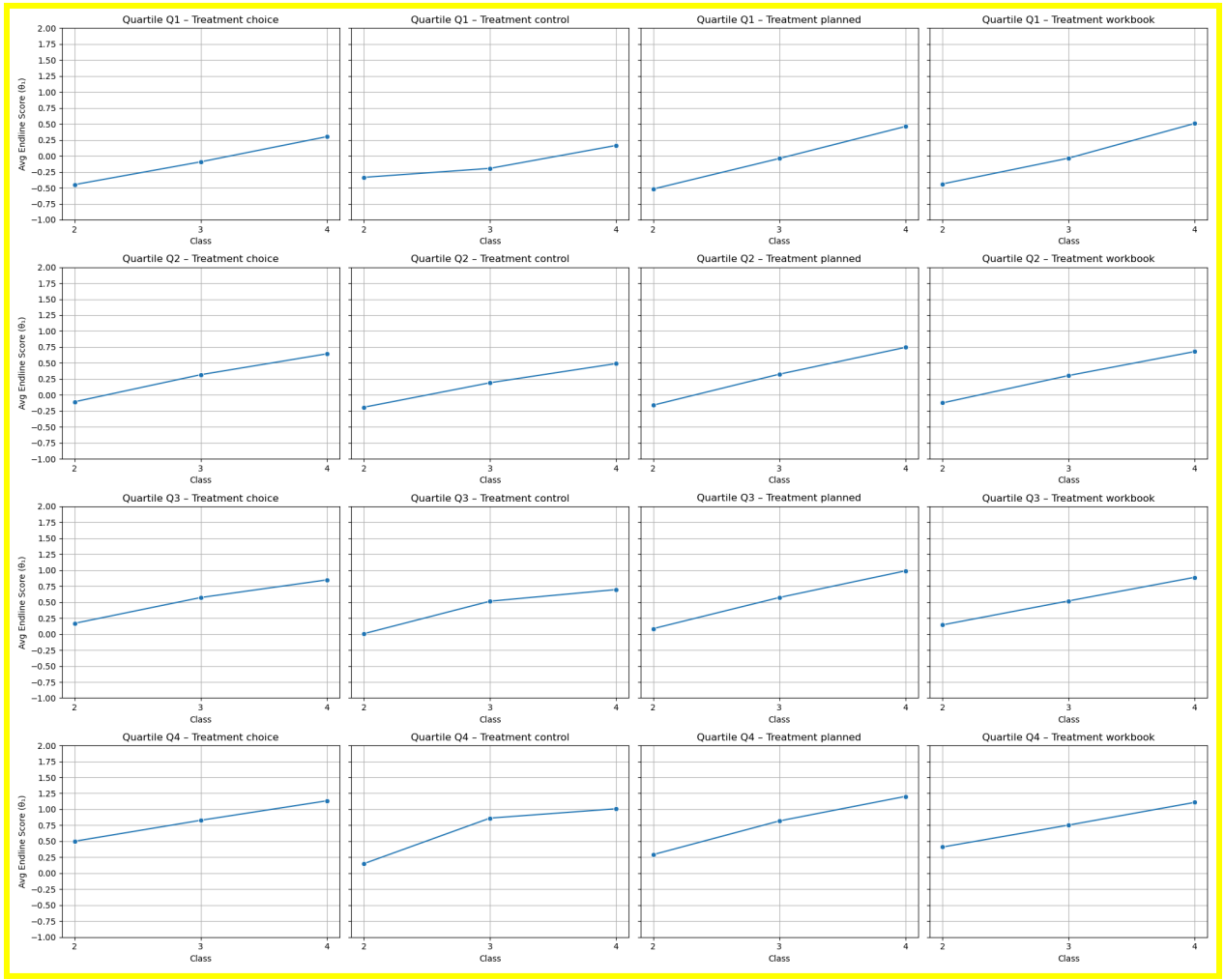
All the classes follow the same trajectory from quartile 1 to quartile 4 . All of them are parallel. This indicates the quartile distribution across each class and treatment . The only difference is with control.
The consistent patterns in the treatment groups suggest that the interventions help create more predictable and equitable learning progress across baseline ability and grade level. In contrast, the Control group's more erratic pattern may reflect uneven instructional quality or variation in implementation, indicating that without intervention, student outcomes are more sensitive to class-level or contextual differences.



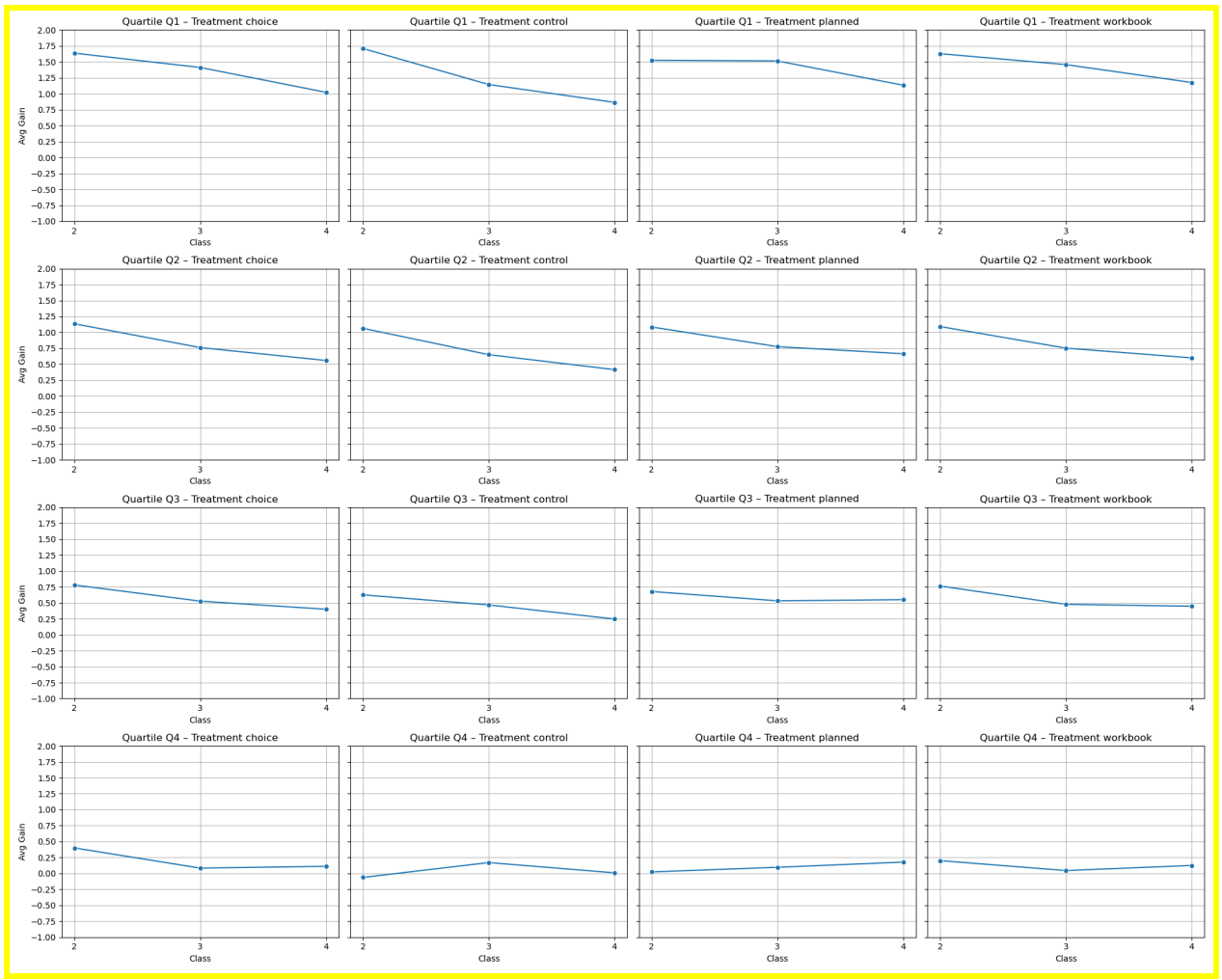
The control line moves from concave to convex and not other treatments. This again shows : The control group shows inconsistent progression — maybe higher-class students benefit more in lower quartiles, but by Q4, the marginal returns reduce (possibly due to saturation or ceiling effects).
Treatments provide more stable gains across class levels and quartiles, possibly indicating they:

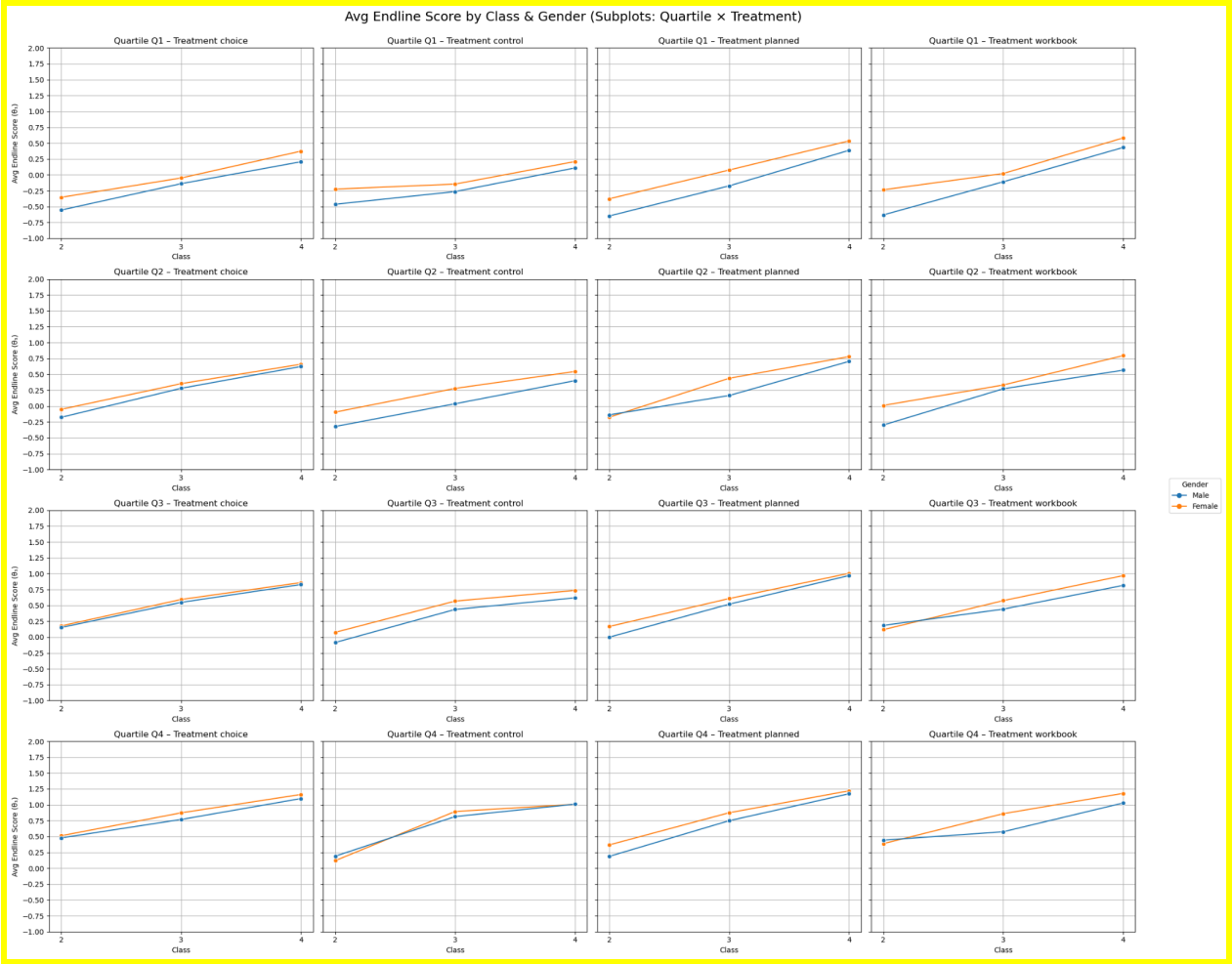
- Reduce disparities in learning gains across class levels.
- Are more systematically effective.





Control shows the most change in line . Not consistent across quartiles


Quartile Q1 – Treatment choice
Quartile Q1 – Treatment control
Quartile Q1 – Treatment planned
Quartile Q1 – Treatment workbook
Quartile Q2 – Treatment choice
Quartile Q2 – Treatment control
Quartile Q2 – Treatment planned
Quartile Q2 – Treatment workbook
Quartile Q3 – Treatment choice
Quartile Q3 – Treatment control
Quartile Q3 – Treatment planned
Quartile Q3 – Treatment workbook
Quartile Q4 – Treatment choice
Quartile Q4 – Treatment control
Quartile Q4 – Treatment planned
Quartile Q4 – Treatment workbook
Avg Gain
Class



I see that males are often disadvantaged however the treatment where the male and females have similar scores is choice

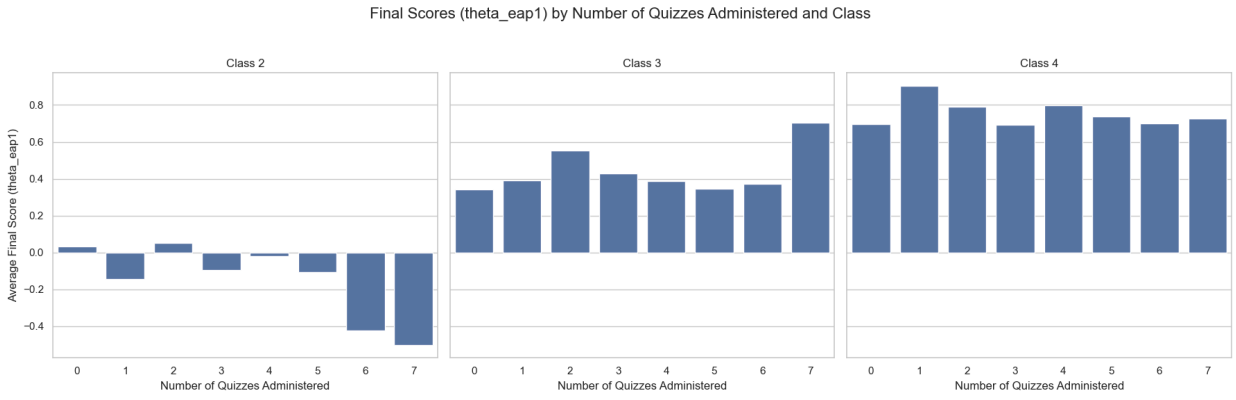
Our data analysis reveals that male students tend to underperform relative to female students across most treatment arms. However, in the TIP Choice version, this gender gap appears to narrow significantly. This may be attributed to the program's unique flexibility, allowing teachers to adapt pedagogy and materials in real-time using monthly performance data. Unlike more rigid structures, TIP Choice empowers educators to address classroom-level disparities — including those rooted in gender — making it a promising model for equitable learning outcomes(This is later also confirmed by Heterogeneity Analysis).

Teacher Compliance Data(Math)(Link)

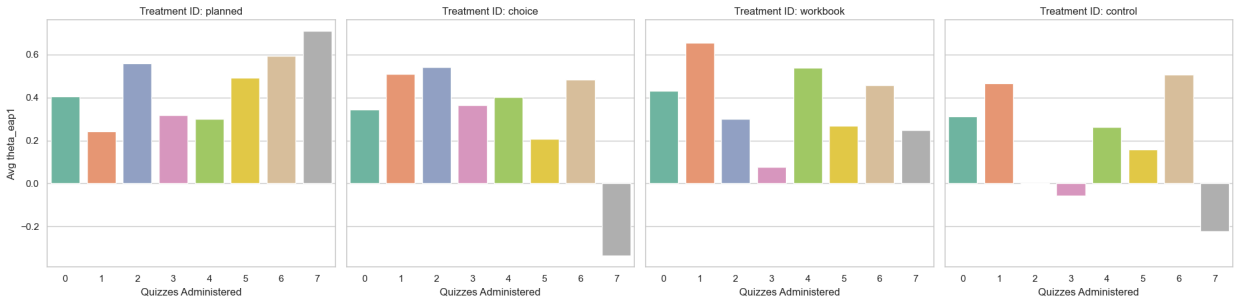
curric_treat_id	avg_participation	avg_total_students	total_quizzes

0	choice	0.647620	34.218750	3.448462
1	control	0.565936	33.602478	2.169400
2	planned	0.636487	33.940662	3.294281
3	workbook	0.577888	34.947178	1.982145

The Choice and Planned groups had the highest number of quizzes administered. Without knowing the program's structure, it is difficult to determine causality. If quizzes were an integral part of the program, any change in performance could be attributed to the treatment itself. However, if quizzes were not embedded in the program, improved performance might instead be explained by the increased monitoring and engagement that frequent quizzes provide

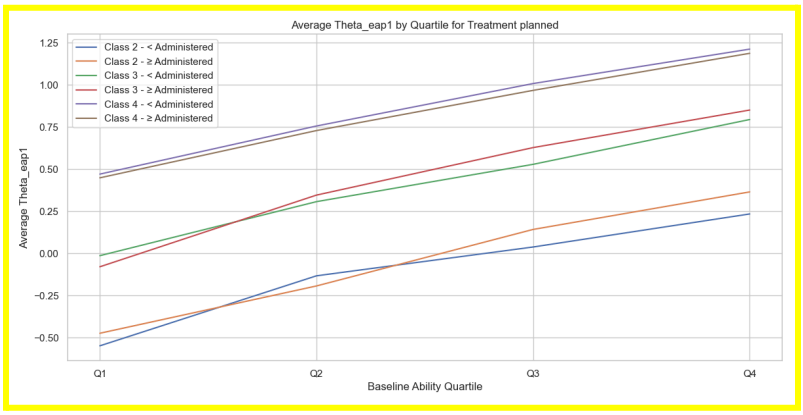


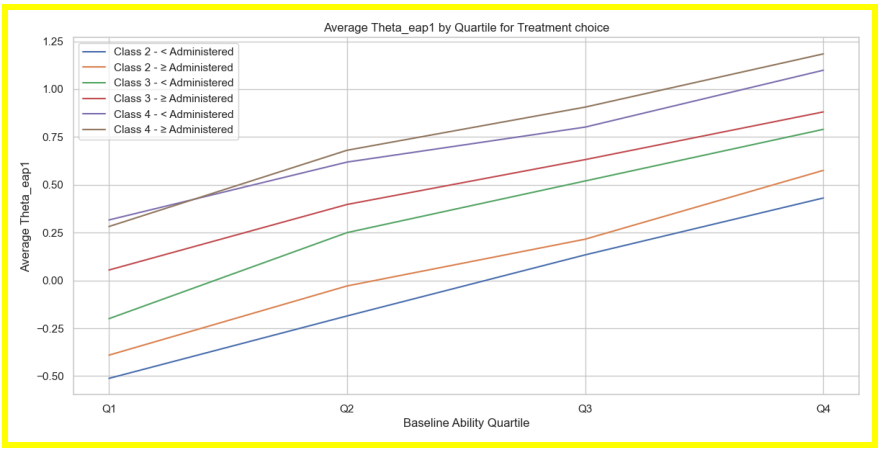
- Class 2: More quizzes tend to negatively impact final scores.
- Class 3: There is an increase in performance as quizzes increase, but the scores fluctuate.
- Class 4: The number of quizzes doesn't seem to have much of an effect on final scores, as the performance remains fairly consistent

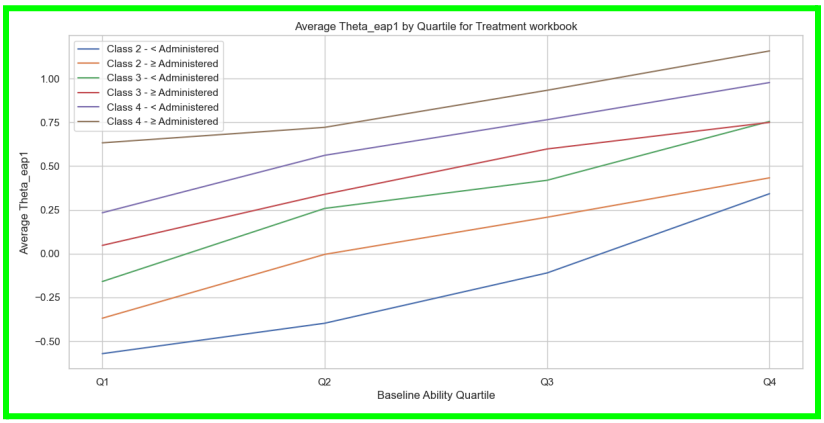


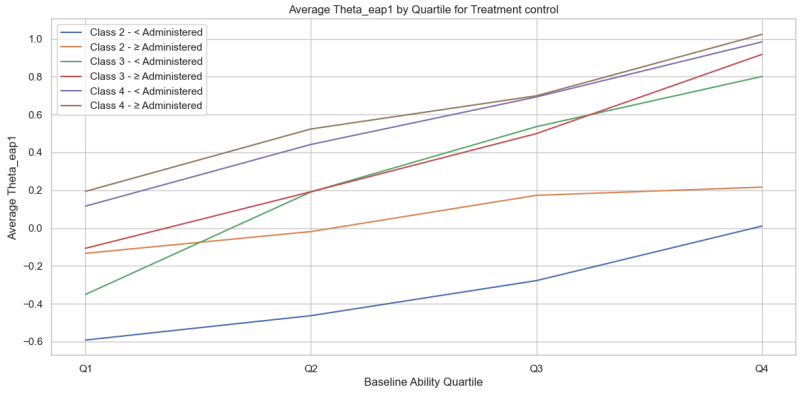
I observed that the number of quizzes administered may play a significant role in the planned treatment, possibly because treatment groups receive structured guidance that encourages regular monitoring through quizzes. This is supported by the trend that endline scores tend to increase with a higher number of quizzes administered, which is not seen in other treatments. Specifically, administering 2 or 6 quizzes appears to consistently yield better performance across all treatment types. However, administering 7 quizzes shows mixed results, with performance varying from positive to negative. This inconsistency warrants further investigation.

Then I looked at the students who responded to fewer quizzes then administered and made line plots of each treatment with lines from each class for those who had less than administered and more than administered.

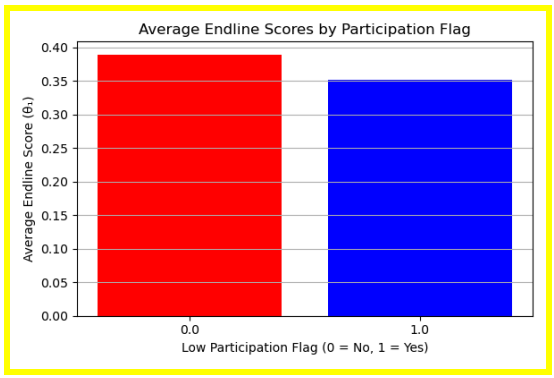




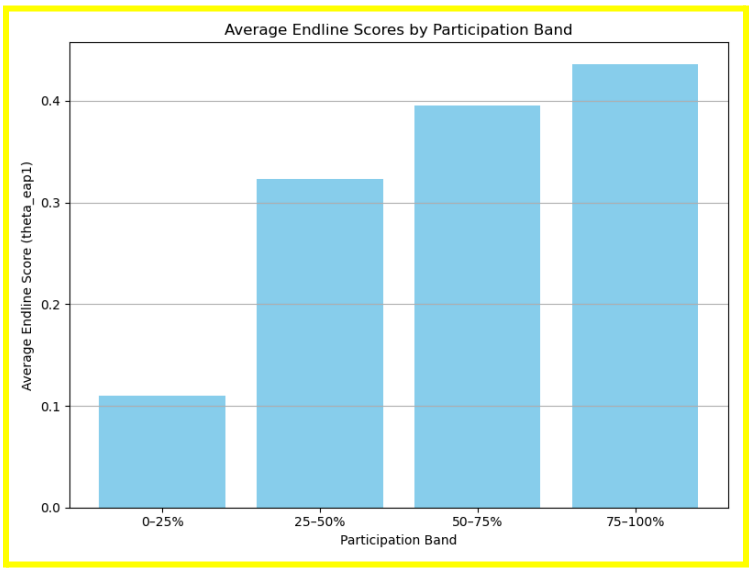




I see the parallel trend again . The same observation I saw previously, but one observation is that the students who attempted all administered performed better than less than administered . It means that those that took all the quizzes performed better .

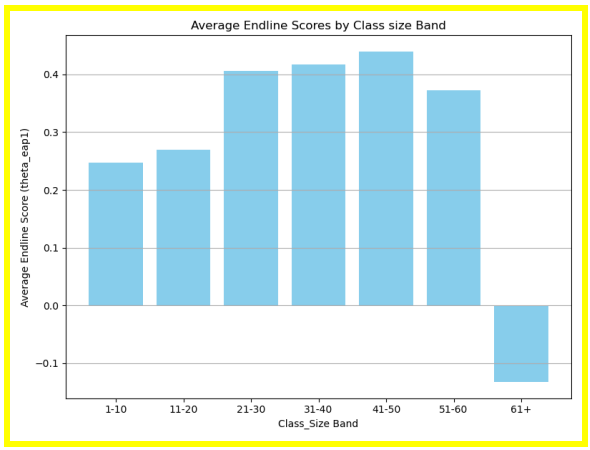


Those flagged with low participation have lower endline scores than higher participation



It is really obvious that students with low participation have lower grades than higher participation . But nothing can be said since the count is smaller for lower participation, so the average may not be representative.

I observed that the class sizes and the participation varied a lot across treatment and classes , which could be some potential environmental factors that influence the endline score because of non standardization . However , running the regression results no significant p-value came for participation or class size



For very huge classes the average score is negative. But nothing can be said again because of lower count in bigger classes

Heterogeneity of Math Data Set(Link)

Using a DRLearner model with Random Forests, I estimated the causal effects (ATEs) of different educational treatments on student post-test scores, controlling for baseline performance, student characteristics, and teacher effects. The analysis compared treatments within subgroups defined by class level, baseline performance quartile, and gender.

- By gender: Males tended to benefit most from choice, females from planned.

- Decision hierarchy: Class level appears most influential, followed by quartile, then gender (for mid-tier groups).

Key Takeaway: The Decision Hierarchy

When assigning treatments, prioritize:

1. Class Level → 2. Performance Quartile → 3. Gender (for mid-tier classes/quartiles).

- Class 4? → Always planned.
- Class 2–3? → Use quartile:
    - Q1 → workbook (all).
    - Q4 → choice (all).
    - Q2–Q3 → planned , choice

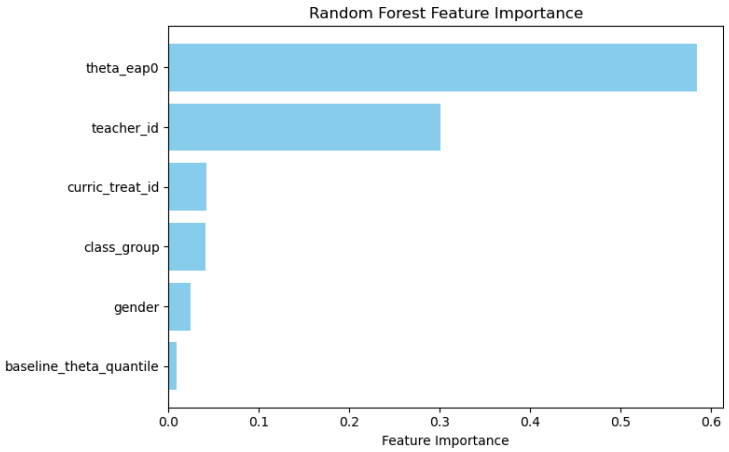
Subgroup regression results were Bonferroni-corrected to account for multiple comparisons, which reduced significance—most strong effects remained only for Class 4 (planned), Quartiles 2–3 (planned, choice), and males (choice).

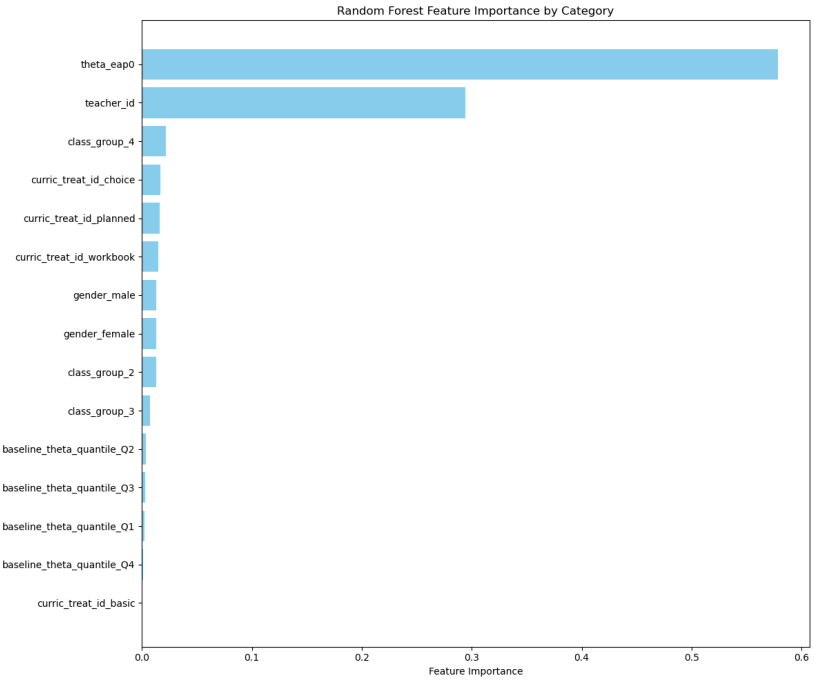
Caution is warranted because subgroup filtering can inflate false positives and the dataset is large, making interaction-term analysis more reliable. These results should be treated as hypothesis-generating, to be confirmed in pre-registered follow-up studies. The patterns nonetheless

align with initial machine learning insights and suggest differentiated treatment strategies by student profile.

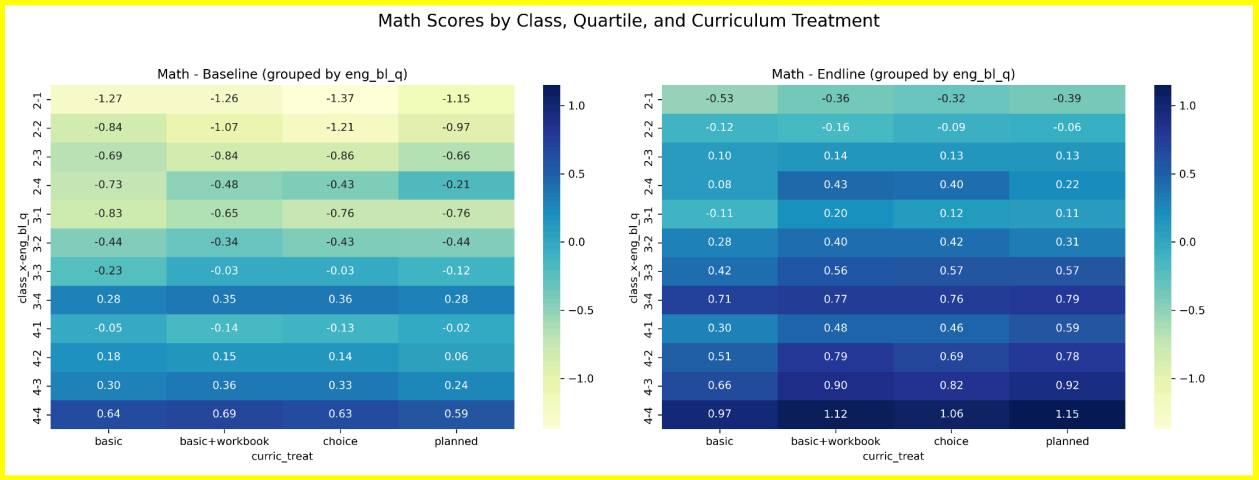
Random Forest Importance of Features(Link)

I cleaned and prepared the dataset by selecting relevant features, handling missing values, and standardizing the gender variable. Then, I one-hot encoded the categorical variables to make the data suitable for modeling. Using a Random Forest Regressor, I trained a model to predict post-treatment learning outcomes (theta_eap1) and visualized the feature importances to identify which variables contributed most to the predictions.



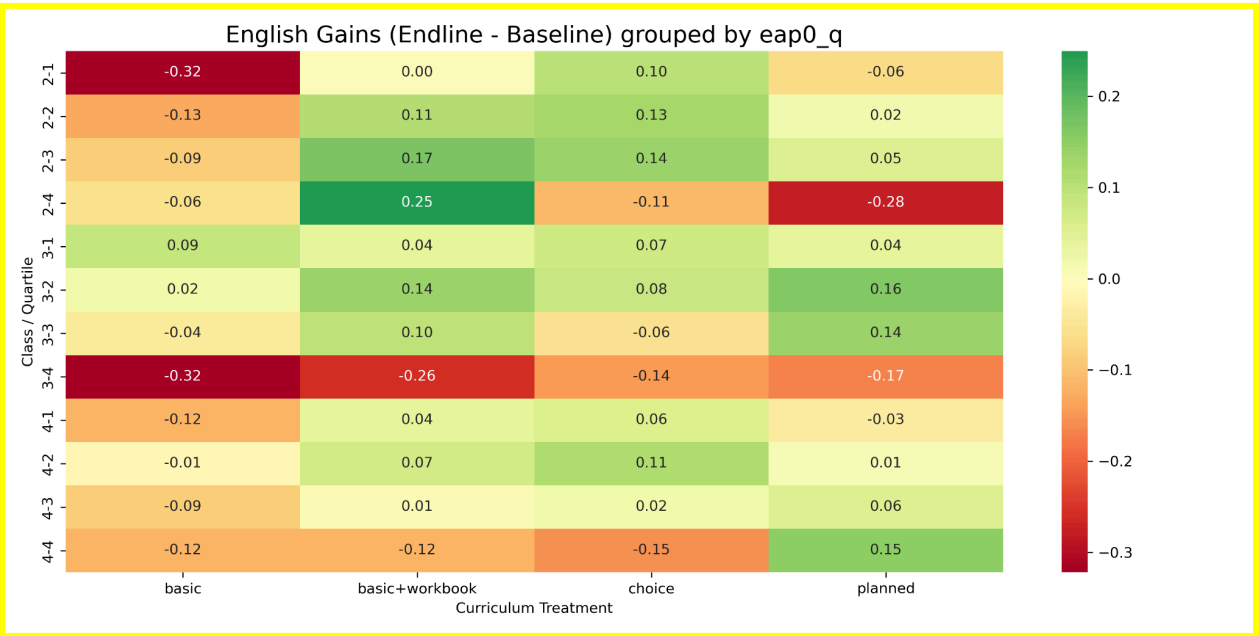


Cross Quartile Heatmaps (Link)



Over here what I noticed which was worth noting was that the quartile differences vanished and the class difference is more pronounced. The quartiles are based on English scores, so they don't meaningfully segment Math ability—hence no quartile differences appear. Class differences persist because systematic factors like teacher quality and curriculum affect all students in a class similarly. This could also mean that the quartiles that were formed in english were not the same in math .**Math quartiles segment ability in a way that is predictive for English, but English quartiles don't segment Math ability well.**

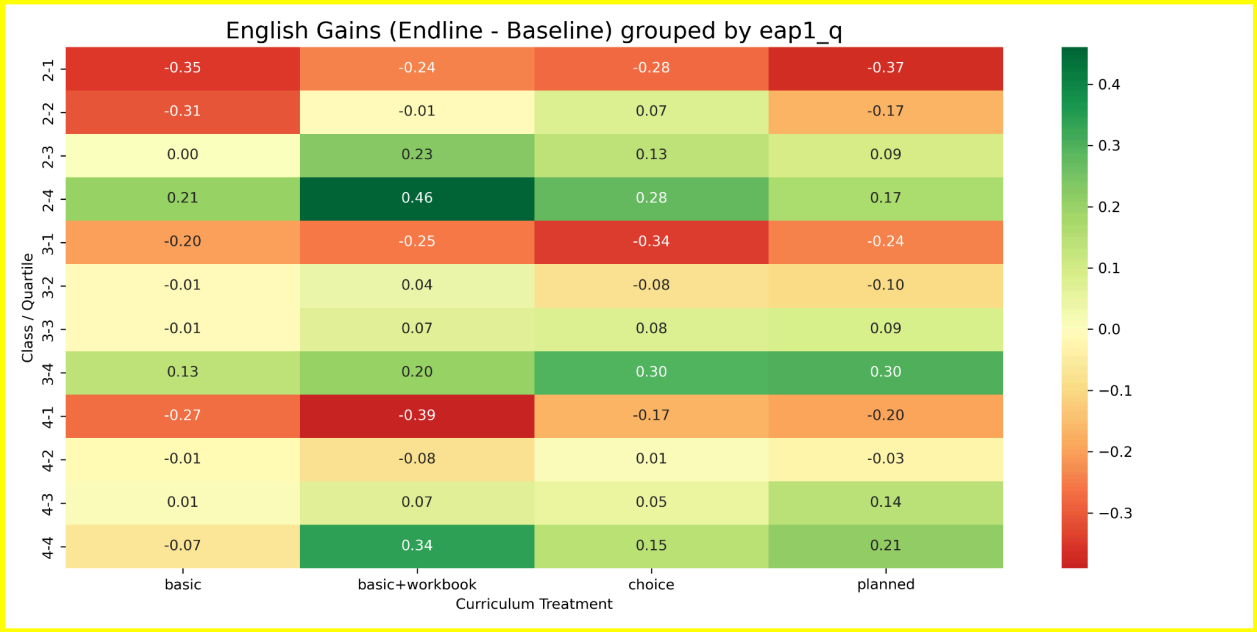
English Gains with Math Baseline Quartiles



I observe no clear trends across quartiles or classes. The most notable pattern is that the Basic group shows the smallest gains, indicated by the orange shading. This suggests that, within Math baseline quartiles, students in the treatment groups achieved greater gains in English than those in the Basic group across all Math quartiles. Essentially, there appears to be a spillover

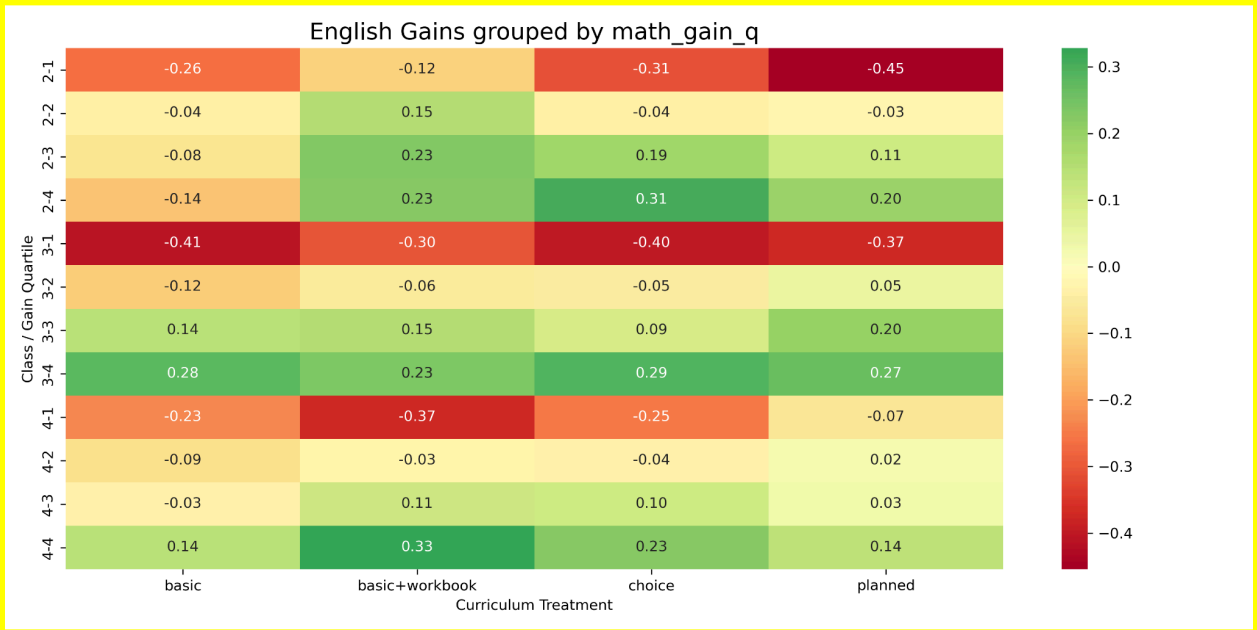
effect of the Math treatments on English performance. However it according to regression analysis it is not significant

English Gains with Math Endline Quartiles



With math endline quartiles . I see the basic treatment difference again. But a major observation is that quartiles show least gains(negative) . This means that students in quartile one of endline math score had the least gain in english as well .Which is expected since those that performed worse in endline would perform worse in english as well and have least gains .

English Gains with Math Gain Quartiles



I see similar trends: the basic group shows the least gain, and students in the first math gain quartile have the lowest gains in English/Urdu (as expected). This suggests a spillover effect of

math improvement on other subjects.However according to th regression Analysis it may not be significant

A similar trend was observed in Urdu , but less noticeable